

Saptarshi Maji, PhD

IPMC-CNRS, Valbonne, France, 06560

+ (33) 074 555 8977

maji@ipmc.cnrs.fr

saps018.github.io

Post-doctoral fellow

Education

- 2018 – 2024 Ph. D. scholar in Department of Biological Sciences (DBS), *IISER Kolkata*.
- 2013 – 2018 BS-MS dual degree in Biological Sciences, *DBS, IISER Kolkata*, CGPA 8.09/10.
- 2013 Higher Secondary Examination, XIIth standard, *WBCHSE Board*, Scored **89%** in science.
- 2011 Secondary Examination, Xth standard, *WBBSE Board*, Scored **95%** in science.

Research Experiences

Post-doctoral experience

- 2024 – Curr Understanding the role of Zinc in regulating the function of a novel keratinocyte differentiation marker, under supervision of *Dr. Bruno Antonny*, IPMC-CNRS, France.

Doctoral thesis

- 2018 – 2024 Functional and Temporal characterisation of Wilson Disease Protein ATP7B at the Lysosomes, under supervision of *Prof. Arnab Gupta*, DBS, IISER Kolkata.

Master's thesis

- 2017 – 2018 Regulatory mechanism of copper transporters hCTR1 and ATP7B for maintaining copper homeostasis, under supervision of *Prof. Arnab Gupta*, DBS, IISER Kolkata.

Summer Internships

- Summer 2016 Over Expression and Purification of Recombinant PCBP2 Protein to Generate Antibody and Study Its Interaction with Coxsackie virus B3 RNA, under supervision of *Prof. Saumitra Das*, Molecular and Cellular Biology, Indian Institute of Science (IISc) Bangalore.
- Summer 2015 Survey of Endosymbiont Infection among Indian Arthropods, under supervision of *Dr. Rhitoban Ray Choudhury*, Department of Biological Sciences, IISER Mohali.
- Summer 2014 Preventing Racemization of Chiral Schiff-base Ligands and Cu(II) Complexes in Solution to synthesize new architecture for chiral recognition, under supervision of *Prof. Manabendra Ray*, Department of Chemistry, Indian Institute of Technology (IIT) Guwahati.

Skills

- Experimental** Experienced with animal cell culture (2D and 3D), genetic engineering and CRISPR technology, general biochemistry and molecular biology techniques, super-resolution microscopy, multi-omics analysis, protein condensate, and lipid biochemistry.
- Software** Proficient in **ImageJ**, **R**, **PYTHON**, **MaxQuant**, **CellProfiler**, **HTML**, and **LaTeX**

Publications

- Maji, B.†, **Maji, S.*†**, Roy, A. N., Singh, R., Yadav, V., Ashok, B. A., Dutta, B., Sarkar, S., Majumder, A., Chakrabarty, M., Dey, S., Dash, T., Pruthwiraj, Pezacki, A. T., Chang, C. J., Das, A., Siddhanta, S., Gupta, A.*; **Copper imbalance in Wilson disease drives dysregulation of the lysosome-mTOR-cholesterol axis in hepatocytes**, (*communicated*) (*Corresponding author)
- Jose, S. M.†, **Maji, S.†**, Paul, M., Singhal, C., Gangappa, S. N., Gupta, A. M., Gupta, A.; **Comparative analysis of key Copper-ATPase domains sheds light on evolutionary divergence in human homologues ATP7A and ATP7B**, *J. Cell Sci.* (2025)
- Maji, S.**, Pirozzi, M., Raturaj, Pandey, R., Ghosh, T., Das, S., Gupta, A.; **Copper-independent lysosomal localization of the Wilson disease protein ATP7B**, *Traffic* (2023)

4. Maji, S., Raturaj, Pandey, R., Das, S., Gupta, A.; **Trans-Golgi Network-Lysosome proximity sites govern trafficking of Wilson disease protein ATP7B in hepatocytes**, *CellBio23*, P2079 (2023)
5. Mondal, B.†, Padhy, A.†, Maji, S.†, Gupta, A., Gupta, S. S.; **Dual stimuli-responsive cross-linked nanoassemblies from an amphiphilic mannose-6-phosphate based tri-block copolymer for lysosomal membrane permeabilization**, *Biomater. Sci.* (2023)
6. Kar, S.†, Sen, S.†, Maji, S., Saraf, D., Raturaj, Paul, R., Dutt, S., Mondal, B., Rodriguez-Boulán, E., Schreiner, R., Sengupta, D., Gupta, A.; **Copper(II) import and reduction are dependent on His-Met clusters in the extracellular amino terminus of human copper transporter-1**, *J. Biol. Chem.* (2022)
7. Das, S., Maji, S.†, Raturaj†, Bhattacharya, I., Saha, T., Naskar, N., Gupta, A.; **Retromer retrieves the Wilson disease protein ATP7B from endolysosomes in a copper-dependent manner**, *J. Cell Sci.* (2020)
8. Mandal, T.†, Kar, S.†, Maji, S.†, Sen, S.†, Gupta, A.; **Structural and Functional Diversity Among the Members of CTR, the Membrane Copper Transporter Family**, *J. Membr. Biol.* (2020)
9. Chatterjee, S. K., Kar, S., Nikte, S., Dash, T., Ghosh, T., Paul, M., Maji, S., Sengupta, D., Sinha, B., Gupta, A.; **Non-canonical regulation of the plasma membrane copper transporter CTR1 through modulation of membrane mechanical properties**, *MBoC* (2026)
10. Pandey, R., Roy, A. N., Sarkar, S., Rohman, R., Chakraborty, K., Bargakshatriya, R., Pandey, S., Pruthwiraj, Bhattacharya, D., Kumar, S., Maji, S., Pezacki, A. T., Pramanik, S. K., Chang, C. J., Bhattacharjee, A., Sengupta, N., Das, A., Gupta, A.; **Repurposing Melatonin in dual-mode for Wilson disease therapy as a Copper Chelator and an antioxidant agent**, *Redox Biol.* (2025)
11. Paul, R., Chakrabarty, A., Samanta, S., Dey, S., Pandey, R., Maji, S., Pezacki, A. T., Chang, C. J., Datta, R., Gupta, A.; **Leishmania major-induced alteration of host cellular and systemic copper homeostasis drives the fate of infection**, *Commun. Biol.* (2024)
12. Raturaj, Mishra, M., Saha, S., Maji, S., Rodriguez-Boulán, E., Schreiner, R., Gupta, A.; **Regulation of apico-basolateral trafficking polarity of homologous Copper-ATPases ATP7A and ATP7B**, *J. Cell Sci.* (2023)
13. Chakraborty, K., Das, S., Pal, A., Maji, S., Rai, B., Gupta, A., Bhattacharjee, A.; **Wilson disease-causing mutations in the carboxyl terminus of ATP7B regulates its localization and Golgi exit selectively in the unpolarized cells**, *Metallomics* (2023)
14. Das, S., Mohammed, A., Mandal, T., Maji, S., Verma, J., Raturaj, Gupta, A.; **Polarized trafficking and copper transport activity of ATP7B: A mutational approach to establish genotype–phenotype correlation in Wilson disease**, *Hum. Mutat.* (2022)
15. Paul, R., Banerjee, S., Sen, S., Dubey, P., Maji, S., Bachhawat, A. K., Datta, R., Gupta, A.; **A novel leishmanial copper P-type ATPase plays a vital role in parasite infection and intracellular survival**, *J. Biol. Chem.* (2021)
16. Maji, M., Acharya, S., Maji, S., Purkait, K., Gupta, A., Mukherjee, A.; **Differences in Stability, Cytotoxicity, and Mechanism of Action of Ru(II) and Pt(II) Complexes of a Bidentate N₂O Donor Ligand**, *Inorg. Chem.* (2020)

†equal contribution

Conference Preceedings

- 2024 Invited as **guest speaker** in **Amity Institute of Biotechnology, Amity University, Kolkata, India**
- 2023 Selected for **poster presentation** in **Cell Bio 23** organised by **ASCB|EMBO, Boston, USA**
 Selected for **poster presentation** in **OBMT 2023** conferenc, **NBRC, India**
 Selected for **oral presentation** in **Organelar Channels and Transporters Gordon Research Conference 2023, Barcelona, Spain**
 Selected for **poster presentation** in **Frontiers in Modern Biology (FiMB), IISER Kolkata**
- 2022 Selected for **poster presentation** in **91st annual meeting of Society of Biological Chemists**
 Selected for **oral presentation** in **Society of Biological Chemists (I) Kolkata chapter**
- 2021 Attended **Molecular Membrane Biology meeting, Online**
- 2020 Selected for **International Symposium on Cell Surface Macromolecules, IISER Pune**

- 2019 Selected for **Multiscale Simulation & Mathematical Modelling of Complex Biological Systems** conference and pre-conference workshop, JNU
- 2018 Selected for **poster presentation** in **Frontiers in Modern Biology (FiMB)**, IISER Kolkata
- 2017 Attended **Advances in Biological Sciences**, IISER Kolkata
- 2015 Attended **Frontiers in Modern Biology (FiMB)**, IISER Kolkata
- 2013 Attended **SPIE winter school on bio-photonics**, IISER Kolkata
- 2011 Selected in **DST-INSPIRE science camp internship** during high school

Fellowships and Awards

- 2026 **Seal of Excellence**, from **Marie Skłodowska-Curie Actions**, European Commission
- 2023 **International Travel Scholarship** by **SCIENCE & ENGINEERING RESEARCH BOARD (SERB)**
International Travel Scholarship by **Research and Development**, IISER Kolkata
Camillo Golgi best poster award in the **Organelle Biogenesis & Membrane Trafficking** conference (OBMT) 2023, NBRC, India
Gold Medal for IISER Kolkata Team in the **iGEM Grand Jamboree 2023**, Paris expo, France
- 2020-2023 **Joint CSIR-UGC NET SRF scholarship** by Govt. of India
- 2018-2020 **Joint CSIR-UGC NET JRF scholarship** by Govt. of India
- 2013-2018 **INSPIRE Fellowship** by Department of **Science and Technology**, Govt. of India

Teaching Experiences

- 2018-2024 Mentored several summer interns, undergraduate and fresh graduate students during PhD tenure
- 2023 **Student mentor** of **iGEM 2023 IISER Kolkata Undergrad Team**
- June, '22 **Module Instructor of Microscopy and Image analysis** for **Karyashala (Workshop) on advanced bio-analytical methods and applications (BAW-2022)**
- Autumn, '22 **Teaching Assistant** for 3rd year BSMS Students for the **Cell Biology and Imaging Laboratory**
- Autumn, '19 **Teaching Assistant** for 4th year BSMS Students for the **Advanced Physiology** course
- Autumn, '17 **Teaching Assistant** for 1st year BSMS Students for the **Basic Life Sciences** course

Professional activities

1. Reviewed for **Apoptosis**, Springer Nature
2. Reviewed for **Cell Biology and Toxicology**, Springer Nature
3. Reviewed for **Bio-protocol Journal**
4. Member of **American Society for Cell Biology (ASCB)**
5. Official Nominator for **VinFuture Prize 2026**, Vietnam

Extracurricular Activities

1. Member of **International Relation and Outreach (IRO)**, IISER Kolkata, India.
2. **Guest Editor** and **content writer** of '**Cogito137**', science communication magazine of IISER Kolkata, India.
3. **Founder member** of '**Art for a Cause**', an NGO that provided aid to Covid affected people
4. **Organiser** and **Volunteer** of several conferences in **IISER Kolkata**
5. **Event photographer** for FIMB 2018, IISM (Inter IISER Sport Meet), and IICM (Inter IISER Cultural Meet).
6. Write poems and short stories in Bengali. Acted as **editor** in several literary magazines.
7. Also interested in painting, photography, traveling, and trekking

For Reference

1. Prof. Arnab Gupta, DBS, IISER Kolkata, India, email: arnab.gupta@iiserkol.ac.in
2. Dr. Bruno Antonny, IPMC-CNRS, France, email: antonny@ipmc.cnrs.fr
3. Prof. Bidisha Sinha, DBS, IISER Kolkata, India. email: bidisha.sinha@iiserkol.ac.in
4. Prof. Rupak Datta, DBS, IISER Kolkata, India. email: rupakdatta@iiserkol.ac.in